

● HARD

● GEOMETRY AND TRIGONOMETRY

Geometry and Trigonometry

11

10 questions · ~15 min

Q1

GEOMETRY AND TRIGONOMETRY

In triangles LMN and RST , angles L and R each have measure 60° , $LN = 10$, and $RT = 30$. Which additional piece of information is sufficient to prove that triangle LMN is similar to triangle RST ?

- A. $MN = 7$ and $ST = 7$
- B. $MN = 7$ and $ST = 21$
- C. The measures of angles M and S are 70° and 60° , respectively.
- D. The measures of angles M and T are 70° and 50° , respectively.

A right circular cone has a volume of $71,148\pi$ cubic centimeters and the area of its base is $5,929\pi$ square centimeters. What is the slant height, in centimeters, of this cone?

- A. 12
- B. 36
- C. 77
- D. 85

A circle in the xy -plane has its center at $-4, -6$. Line k is tangent to this circle at the point $-7, -7$. What is the slope of line k ?

A. -3

B. $-\frac{1}{3}$

C. $\frac{1}{3}$

D. 3

$$RS = 20$$

$$ST = 48$$

$$TR = 52$$

The side lengths of right triangle RST are given. Triangle RST is similar to triangle UVW , where S corresponds to V and T corresponds to W . What is the value of $\tan W$?

A. $\frac{5}{13}$

B. $\frac{5}{12}$

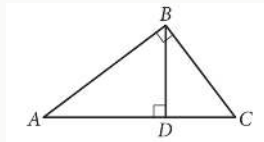
C. $\frac{12}{13}$

D. $\frac{12}{5}$

Q5

GRID-IN

GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

In the figure above, $BD = 6$ and $AD = 8$. What is the length of $\overline{DC}$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The volume of right circular cylinder A is 22 cubic centimeters. What is the volume, in cubic centimeters, of a right circular cylinder with twice the radius and half the height of cylinder A?

- A. 11
- B. 22
- C. 44
- D. 66

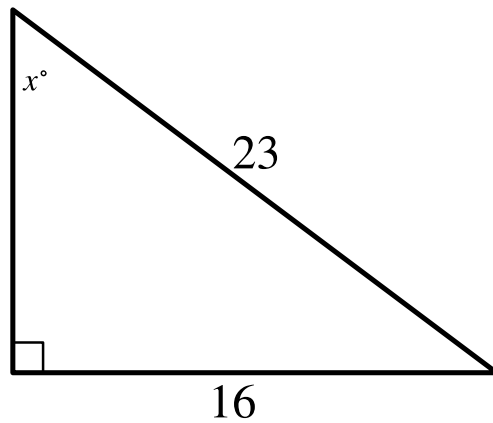
In the xy -plane, a circle has center C with coordinates h, k . Points A and B lie on the circle. Point A has coordinates $h + 1, k + \sqrt{102}$, and $\angle ACB$ is a right angle. What is the length of $\overline{AB}$?

A. $\sqrt{206}$

B. $2\sqrt{102}$

C. $103\sqrt{2}$

D. $103\sqrt{3}$



Note: Figure not drawn to scale.

- One angle is a right angle.
- The measure of a second angle is x° .
- The length of the hypotenuse is 23.
- The length of the leg opposite the angle with measure x° is 16.
- Note: Figure not drawn to scale.

In the triangle shown, what is the value of $\sin x^\circ$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

A manufacturer determined that right cylindrical containers with a height that is 4 inches longer than the radius offer the optimal number of containers to be displayed on a shelf. Which of the following expresses the volume, V , in cubic inches, of such containers, where r is the radius, in inches?

- A. $V = 4\pi r^3$
- B. $V = \pi(2r)^3$
- C. $V = \pi r^2 + 4\pi r$
- D. $V = \pi r^3 + 4\pi r^2$

Q10

GRID-IN

GEOMETRY AND TRIGONOMETRY

Circle A in the xy -plane has the equation $x + 5^2 + y - 5^2 = 25$. Circle B has the same center as circle A. The radius of circle B is two times the radius of circle A. The equation defining circle B in the xy -plane is $x + 5^2 + y - 5^2 = k$, where k is a constant. What is the value of k ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.